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JEE Main Physics Practice Sheet — DPP-042

Topic: Conservative and Non-Conservative Forces — Distinguishing Features & Closed-Loop Work

Time Allowed: 60 Minutes | **Max Marks:** 80 | **Marking Scheme:** +4, -1

Q1. A force field $\vec{F}$ is defined to be conservative if and only if:

(A) Work done over any closed path is zero ($\oint \vec{F} \cdot d\vec{r} = 0$)
(B) Work done between two endpoints is independent of the trajectory taken
(C) The curl of the force field is identically zero ($\vec{\nabla} \times \vec{F} = \vec{0}$)
(D) All of the above

Q2. Which of the following force fields is NON-CONSERVATIVE?

(A) $\vec{F} = (2xy\hat{i} + x^2\hat{j})$ N
(B) $\vec{F} = (y\hat{i} + x\hat{j})$ N
(C) $\vec{F} = (y\hat{i} - x\hat{j})$ N
(D) $\vec{F} = (3x^2\hat{i} + 4y^3\hat{j})$ N

Q3. A block of mass m is dragged along a closed square path of side a on a rough horizontal floor with kinetic friction coefficient μ_k. The net work done by the frictional force over the closed loop is:

(A) Zero
(B) $-4\mu_k mga$
(C) $+4\mu_k mga$
(D) $-2\mu_k mga$

Q4. For which of the following forces can a meaningful, single-valued potential energy function $U(\vec{r})$ be uniquely defined?

(A) Dynamic viscous drag force on a sphere
(B) Kinetic friction force on a sliding block
(C) Gravitational electrostatic central force
(D) Air resistance proportional to velocity squared

Q5. A particle moves along a circular path of radius R in the xy-plane centered at the origin under the force $\vec{F} = c(y\hat{i} - x\hat{j})$, where c is a constant. The work done by this force in one complete counterclockwise revolution is:

(A) Zero
(B) $-2\pi cR^2$
(C) $+2\pi cR^2$
(D) πcR^2

Q6. For a force $\vec{F} = (ax\hat{i} + by\hat{j})$ N, the work done in moving a particle from $(0,0)$ to $(1\text{ m}, 1\text{ m})$ along path 1 ($y = x$) is W_1, and along path 2 ($y = x^2$) is W_2. The relation between W_1 and W_2 is:

(A) $W_1 > W_2$
(B) $W_1 < W_2$
(C) $W_1 = W_2 = \dfrac{a+b}{2}$
(D) $W_1 = 2W_2$

Q7. A body of mass m is taken from point A to point B in a conservative gravitational field along three different paths of lengths s_1, s_2, s_3 ($s_1 < s_2 < s_3$). If W_1, W_2, W_3 are the work done by gravity along these paths respectively, then:

(A) $W_1 = W_2 = W_3$
(B) $W_3 > W_2 > W_1$
(C) $W_1 > W_2 > W_3$
(D) $W_1 + W_3 = 2W_2$ only if $s_2 = (s_1 + s_3)/2$

Q8. If a force $\vec{F} = (f_x\hat{i} + f_y\hat{j})$ is conservative in 2D space, the partial derivatives must satisfy:

(A) $\dfrac{\partial f_x}{\partial x} = \dfrac{\partial f_y}{\partial y}$
(B) $\dfrac{\partial f_x}{\partial y} = \dfrac{\partial f_y}{\partial x}$
(C) $\dfrac{\partial f_x}{\partial y} = -\dfrac{\partial f_y}{\partial x}$
(D) $\dfrac{\partial f_x}{\partial x} + \dfrac{\partial f_y}{\partial y} = 0$

Q9. Which of the following statements is INCORRECT regarding non-conservative forces?

(A) Work done over a closed path is always non-zero
(B) Mechanical energy is always strictly conserved in their presence
(C) They often dissipate mechanical energy into thermal energy or internal energy
(D) Work done depends on the specific path taken between two points

Q10. A particle moves from point $A(1,2)$ to point $B(3,4)$ under a force $\vec{F} = (2x\hat{i} + 2y\hat{j})$ N. The work done by $\vec{F}$ is:

(A) 20 J
(B) 12 J
(C) 8 J
(D) 24 J

Q11. Consider a central force $\vec{F} = f(r)\hat{r}$ where r is the radial distance. Such a central force is:

(A) Non-conservative if $f(r) \propto 1/r^3$
(B) Always conservative for any continuous function $f(r)$
(C) Conservative only if $f(r) \propto 1/r^2$
(D) Non-conservative because work depends on angle θ

Q12. A particle of mass m is moved slowly along a closed loop in a vertical plane in the presence of uniform gravity and air resistance. The net work done by all forces in one complete cycle is:

(A) Strictly negative
(B) Zero
(C) Strictly positive
(D) $+mgh$

Q13. A force $\vec{F} = (ay^2\hat{i} + 2bxy\hat{j})$ N acts on a particle. For what ratio of a/b is this force conservative?

(A) $a/b = 1$
(B) $a/b = 2$
(C) $a/b = 1/2$
(D) $a/b = -1$

Q14. A block of mass $m = 2$ kg is pulled at constant speed along a rough semi-circular path of radius $R = 2$ m from $(-R, 0)$ to $(+R, 0)$ ($\mu_k = 0.5$). The work done by friction is ($g = 10$ m/s^2):

(A) -20π J
(B) -40 J
(C) Zero
(D) $+20\pi$ J

Q15. If the work done by a force field along any closed curve C is $\oint_C \vec{F} \cdot d\vec{r} = 0$, by Stokes' Theorem this implies:

(A) $\vec{\nabla} \cdot \vec{F} = 0$
(B) $\vec{\nabla} \times \vec{F} = \vec{0}$
(C) $\vec{\nabla}^2 \vec{F} = \vec{0}$
(D) $\dfrac{\partial \vec{F}}{\partial t} = \vec{0}$

Q16. A particle moves in the xy-plane from $(0, 0)$ to $(1\text{ m}, 1\text{ m})$ under the force $\vec{F} = (x^2\hat{i} + y^2\hat{j})$ N. The work done is:

(A) $\frac{1}{3}$ J
(B) $\frac{2}{3}$ J
(C) 1 J
(D) $\frac{4}{3}$ J

Q17. An object moves under a force $\vec{F} = (3x^2y\hat{i} + x^3\hat{j})$ N along the parabolic path $y = x^2$ from $(0, 0)$ to $(2\text{ m}, 4\text{ m})$. The work done by the force is:

(A) 16 J
(B) 32 J
(C) 64 J
(D) 24 J

Q18. Which of the following is an example of a non-conservative force that can do POSITIVE work on a body?

(A) Static friction accelerating a box on a speeding truck bed
(B) Kinetic friction on a block sliding on a stationary table
(C) Air resistance on a descending skydiver
(D) Viscous drag on a settling colloid

Q19. When a non-conservative force does negative work on a system, the total mechanical energy $E = K + U$ of the system:

(A) Increases
(B) Remains constant
(C) Decreases
(D) Oscillates sinusoidally

Q20. A force $\vec{F} = (z\hat{i} + x\hat{j} + y\hat{k})$ N acts on a particle. The curl $\vec{\nabla} \times \vec{F}$ is:

(A) $\vec{0}$ (Conservative)
(B) $(\hat{i} + \hat{j} + \hat{k})$ (Non-conservative)
(C) $(-\hat{i} - \hat{j} - \hat{k})$ (Non-conservative)
(D) $(2\hat{i} - \hat{j})$ (Non-conservative)

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-042)

Conservative and Non-Conservative Forces — Distinguishing Features & Closed-Loop Work

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	D	6	C	11	B	16	B
2	C	7	A	12	A	17	B
3	B	8	B	13	A	18	A
4	C	9	B	14	A	19	C
5	B	10	A	15	B	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (D)
A force field is conservative if: 1. $\oint \vec{F}\cdot d\vec{r} = 0$ over any closed loop. 2. $\int_A^B \vec{F}\cdot d\vec{r}$ depends solely on endpoints A and B. 3. $\vec{\nabla}\times\vec{F} = \vec{0}$ (curl-free). All three conditions are mathematically equivalent.

Sol 2. Option (C)
For $\vec{F} = (y\hat{\imath} - x\hat{\jmath})$:

$$\vec{\nabla}\times\vec{F} = \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\hat{k} = (-1-1)\hat{k} = -2\hat{k} \neq \vec{0}$$

Since the curl is non-zero, this force is non-conservative. (All other options have zero curl).

Sol 3. Option (B)
Kinetic friction always opposes instantaneous displacement along each side of the square:

$$W_{\text{side}} = -f_k a = -\mu_k mga$$

For all four sides:

$$W_{\text{total}} = 4(-\mu_k mga) = -4\mu_k mga$$

Sol 4. Option (C)
A single-valued potential energy function $U(\vec{r})$ can only be defined for conservative force fields ($\vec{\nabla}\times\vec{F} = \vec{0}$) such as gravitational and electrostatic central fields.

Sol 5. Option (B)
In polar coordinates: $x = R\cos\theta, y = R\sin\theta$, $dx = -R\sin\theta\, d\theta, dy = R\cos\theta\, d\theta$.

$$dW = c(y\,dx - x\,dy) = c[R\sin\theta(-R\sin\theta) - R\cos\theta(R\cos\theta)]d\theta = -cR^2\,d\theta$$

$$W = \int_0^{2\pi} -cR^2\,d\theta = -2\pi cR^2$$

Sol 6. Option (C)
The force is conservative because $dW = ax\,dx + by\,dy = d\left(\frac{1}{2}ax^2 + \frac{1}{2}by^2\right)$.
Work is path-independent:

$$W_1 = W_2 = \left[\frac{1}{2}ax^2 + \frac{1}{2}by^2\right]_{(0,0)}^{(1,1)} = \frac{a+b}{2}$$

Sol 7. Option (A)
Gravity is a conservative force, so the work done depends purely on the vertical displacement between A and B, completely independent of path length ($W_1 = W_2 = W_3$).

Sol 8. Option (B)
The curl condition $\vec{\nabla}\times\vec{F} = \vec{0}$ in 2D reduces to:

$$\frac{\partial f_y}{\partial x} - \frac{\partial f_x}{\partial y} = 0 \implies \frac{\partial f_x}{\partial y} = \frac{\partial f_y}{\partial x}$$

Sol 9. Option (B)
In the presence of non-conservative forces like friction, mechanical energy is converted into heat or other internal forms, so mechanical energy is NOT conserved ($W_{\text{nc}} = \Delta E_{\text{mech}} \neq 0$). Hence statement (B) is incorrect.

Sol 10. Option (A)
$dW = 2x\,dx + 2y\,dy = d(x^2+y^2)$.

$$W = \left[x^2+y^2\right]_{(1,2)}^{(3,4)} = (3^2+4^2) - (1^2+2^2) = 25 - 5 = 20 \text{ J}$$

Sol 11. Option (B)
Any purely central force $\vec{F} = f(r)\hat{r}$ has spherical symmetry and zero curl ($\vec{\nabla}\times(f(r)\hat{r}) = \vec{0}$). Hence, central forces are always conservative.

Sol 12. Option (A)
Work done by gravity over a closed loop is zero ($W_g = 0$). Air resistance always opposes velocity ($dW_{\text{air}} = -F_{\text{air}}ds < 0$), so $W_{\text{air}} < 0$.

$$W_{\text{net}} = W_g + W_{\text{air}} = 0 + (\text{negative}) < 0$$

Sol 13. Option (A)
For a conservative force: $\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x}$.

$$\frac{\partial}{\partial y}(ay^2) = 2ay, \quad \frac{\partial}{\partial x}(2bxy) = 2by$$

$$2ay = 2by \implies a = b \implies \frac{a}{b} = 1$$

Sol 14. Option (A)
Length of semi-circular track $= \pi R = 2\pi$ m.
Kinetic friction force: $f_k = \mu_k mg = 0.5\times 2\times 10 = 10$ N.
Work done by friction:

$$W_f = -f_k(\pi R) = -10(2\pi) = -20\pi \text{ J}$$

Sol 15. Option (B)
By Stokes' Theorem:

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_S (\vec{\nabla}\times\vec{F})\cdot d\vec{A} = 0 \iff \vec{\nabla}\times\vec{F} = \vec{0}$$

Sol 16. Option (B)
$dW = x^2\,dx + y^2\,dy = d\left(\frac{x^3+y^3}{3}\right)$.

$$W = \left[\frac{x^3+y^3}{3}\right]_{(0,0)}^{(1,1)} = \frac{1+1}{3} = \frac{2}{3} \text{ J}$$

Sol 17. Option (B)
$dW = 3x^2y\,dx + x^3\,dy = d(x^3y)$.

$$W = \left[x^3y\right]_{(0,0)}^{(2,4)} = (2^3\times 4) - 0 = 32 \text{ J}$$

Sol 18. Option (A)
Static friction on a block resting on an accelerating truck bed accelerates the block forward in the direction of motion relative to the ground, doing positive work on the block ($W = f_s s > 0$).

Sol 19. Option (C)
By the Work-Energy principle:

$$W_{\text{nc}} = \Delta E_{\text{mech}}$$

If $W_{\text{nc}} < 0$, then $\Delta E_{\text{mech}} < 0$, so the mechanical energy decreases.

Sol 20. Option (B)
Evaluating curl:

$$\vec{\nabla}\times\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \hat{i}(1-0)-\hat{j}(0-1)+\hat{k}(1-0) = (\hat{i}+\hat{j}+\hat{k}) \neq \vec{0}$$

Hence the force is non-conservative.